

Divergence Theorem:

$$\vec{v} = v_1(x_1, x_2) \hat{e}_1 + v_2(x_1, x_2) \hat{e}_2$$

$$\text{Green's theorem} \left\{ \begin{array}{l} \int_R \frac{\partial P}{\partial x} dA = \int_C P n_x ds \\ \int_R \frac{\partial P}{\partial y} dA = \int_C P n_y ds \end{array} \right.$$

$$\Rightarrow \left\{ \begin{array}{l} \int_R \frac{\partial v_1}{\partial x_1} dA = \int_C v_1 n_1 ds \\ \int_R \frac{\partial v_2}{\partial x_2} dA = \int_C v_2 n_2 ds \end{array} \right.$$

$$\int_R \left(\frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} \right) dA = \int_C (v_1 n_1 + v_2 n_2) ds$$

$$\int_R \left(\frac{\partial v_i}{\partial x_i} \right) dA = \int_C (v_i n_i) ds$$

$$\int_R (\text{div } \vec{v}) dA = \int_C (\vec{v} \cdot \vec{n}) ds$$

$$\int_V (\text{div } \vec{v}) dV = \int_S (\vec{v} \cdot \vec{n}) dS$$

$$\alpha (\text{scalar}) : \int_V (\text{div } \alpha \vec{v}) dV = \int_S (\alpha \vec{v} \cdot \vec{n}) dS$$

$$\text{2nd-order tensor:} \int_V (\text{div } \mathbf{T}) dV = \int_S (\mathbf{T} \cdot \vec{n}) dS$$

$$\int_V \frac{\partial T_{ij}}{\partial x_j} dV = \int_S T_{ij} n_j dS$$

3rd-order tensor:

$$\int_V \frac{\partial T_{ijk}}{\partial x_k} dV = \int_S T_{ijk} n_k dS$$

Example 7.2.1

$$f = \int_V (\text{div } T) dV$$

$$f = \int_{S'} t dS' = \int_{S'} T_n dS' = \int_V (\text{div } T) dV \Rightarrow f_i = \int_V \frac{\partial T_{ij}}{\partial x_j} dV$$

Example 7.2.2

$$M = \int_V \left\{ x \times (\text{div } T) + 2t^A \right\} dV$$

$$M = \int_{S'} (x \times t) dS' = \int_{S'} x \times (T_n) dS'$$

$$x_j \times (T_n)_k$$

$$M_i = \int_{S'} x_j \times (T_n)_k dS' = \hat{e}_i \int_{S'} \epsilon_{jki} x_j (T_n)_k dS'$$

$\begin{matrix} \nearrow i \\ k+j \\ \searrow \end{matrix}$

$$= \hat{e}_i \int_{S'} (\epsilon_{ijk} x_j T_{km}) n_m dS' = \hat{e}_i \int_V \frac{\partial (\epsilon_{ijk} x_j T_{km})}{\partial x_m} dV$$

$$= \hat{e}_i \int_V \left\{ \cancel{\frac{\partial \epsilon_{ijk}}{\partial x_m}} x_j T_{km} + \epsilon_{ijk} \cancel{\frac{\partial x_j}{\partial x_m}} T_{km} + \epsilon_{ijk} x_j \frac{\partial T_{km}}{\partial x_m} \right\} dV$$

δ_{jm}

$$= \hat{e}_i \int \left\{ \epsilon_{ijk} T_{kj} + \epsilon_{ijk} x_j \frac{\partial T_{km}}{\partial x_m} \right\} dV$$

↓ swap k and j

$$= \hat{e}_i \int \left\{ \cancel{\epsilon_{ikj}} T_{jk} + \epsilon_{ijk} x_j \frac{\partial T_{km}}{\partial x_m} \right\} dV$$

$-\epsilon_{ijk}$

$$= \hat{e}_i \int \left\{ -\epsilon_{ijk} T_{jk} + \epsilon_{ijk} x_j \frac{\partial T_{km}}{\partial x_m} \right\} dV$$

$$2t^A = -\epsilon_{ijk} T_{jk} \hat{e}_i$$

$$\xrightarrow{\text{blue arrow}} M = \int_V (2t^A + x \times \text{div } T) dV$$

Example 7.2.3

$$P = \int_S t \cdot v \, dS = \int_V \left\{ (\text{div } T) \cdot v + \text{tr}(T^T \nabla v) \right\} dV$$

$$P = \int_S t \cdot v \, dS = \int_S (Tn) \cdot v \, dS = \int_S n \cdot T^T v \, dS = \int_S T^T v \cdot n \, dS$$

$$= \int_V \text{div}(T^T v) \, dV$$

$$\text{div}(T^T v) = \frac{\partial (T_{ij}^T v_j)}{\partial x_i} = \frac{\partial (T_{ji} v_j)}{\partial x_i}$$

$$= \frac{\partial T_{ji}}{\partial x_i} v_j + T_{ji} \frac{\partial v_j}{\partial x_i} = \text{div } T^T \cdot v + \text{tr}(T^T \nabla v)$$

$$P = \int_V \left(\text{div } T^T \cdot v + \text{tr}(T^T \nabla v) \right) dV$$